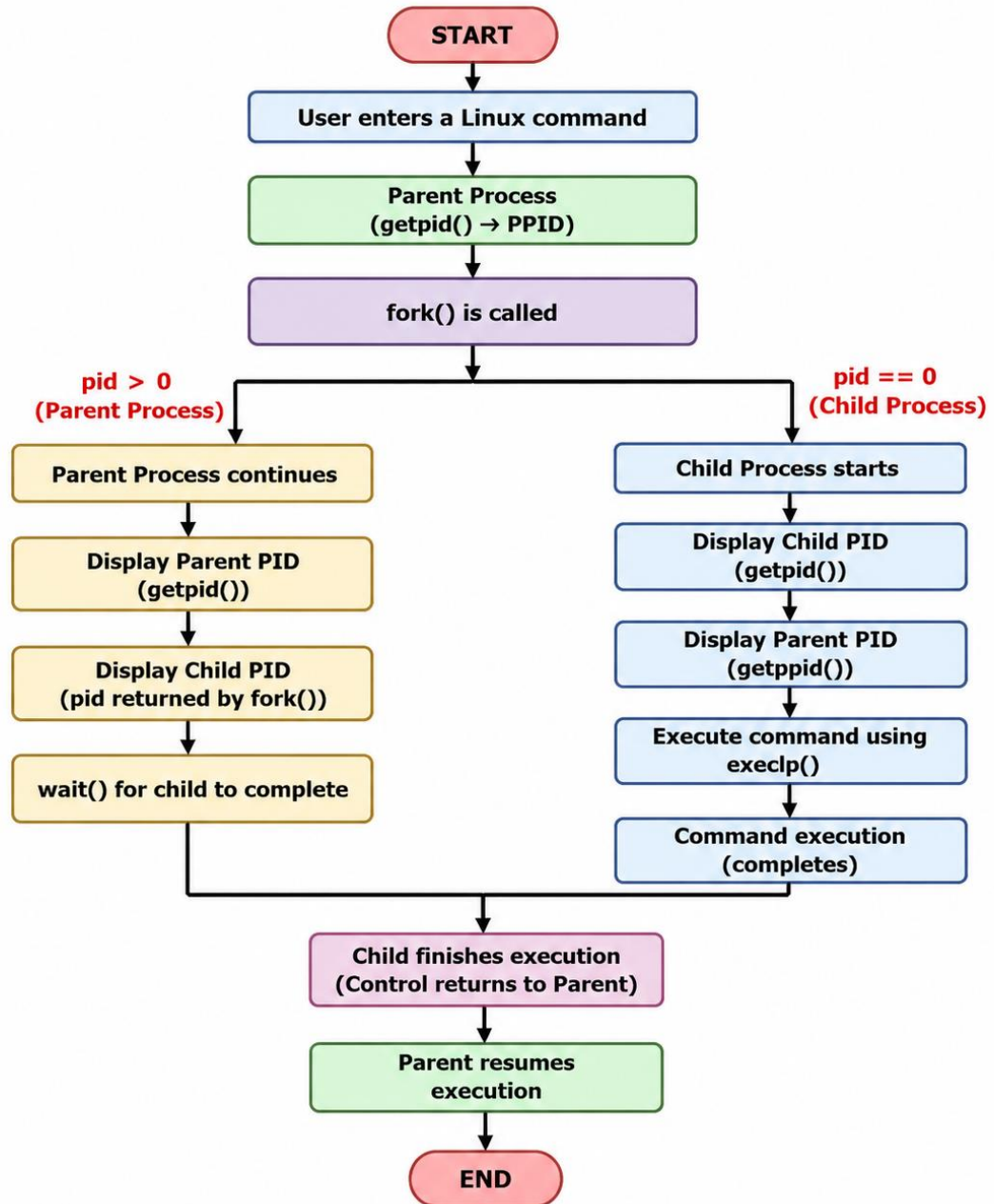


Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) **Write a C code to create a process using fork() system call.**
- b) **Read the user input as shell command.**
- c) **Use a system call exec() to execute the shell command.**
- d) **Use a wait() system call for waiting a parent process.**
- e) **Display the PIDs of parent and child processes.**



Example Output:

Enter a Linux command: ls

Parent Process

Child Process

Parent PID : 67

Child PID : 68

Child PID : 68

Parent PID : 67

Executing command: ls

CE1 FIRST.c File2.c Proces_States.c command_executor filecopy.c
hello.c process_example session_4_a session_4_b.c

CE1.c File1.c First.c Process_States.c command_executor.c fork_example.c.save
my_shell process_example.c session_4_a.c

FIRST File1.txt Proces_States R1 filecopy hello
os-lab process_example.cyes session_4_b

Child process completed.

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if(pid < 0)
    {
        printf("Fork Failed!\n");
        return 1;
    }

    else if(pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Execution Failed!\n");
    }

    else
    {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

^G Help ^O Write Out ^W Where Is ^X Cut ^T Execute ^C Location ^M-U Undo ^M-A Set Mark ^M-J To Bracket ^M-Q Previous ^B Back ^_ Prev Word
^X Exit ^R Read File ^H Replace ^U Paste ^J Justify ^_/ Go To Line ^M-E Redo ^M-G Copy ^Q Where Was ^M-W Next ^F Forward ^_ Next Word

```
root@keerthana: ~/OSSP
root@keerthana:~/OSSP# cd ~/OSSP
root@keerthana:~/OSSP# mkdir ~/OSSP
mkdir: cannot create directory '/root/OSSP': File exists
root@keerthana:~/OSSP#
root@keerthana:~/OSSP# cd ~/OSSP
root@keerthana:~/OSSP# pwd
/root/OSSP
root@keerthana:~/OSSP# nano task1.c
root@keerthana:~/OSSP# gcc task1.c -o task1
root@keerthana:~/OSSP# ./task1
Enter a Linux command: ls

Parent Process
Parent PID : 532

Child Process
Child PID : 533
Executing command: ls

LICENSE  README.md  bin  include  scripts  task1  tests
Makefile  assets    docs  obj      src      task1.c

Child process completed.
root@keerthana:~/OSSP# |
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

1.Uname

```
root@keerthana: ~  
root@keerthana:~# uname  
Linux  
root@keerthana:~# uname -a  
Linux keerthana 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
```

2.Lspu

```
root@keerthana:~# lscpu  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:          39 bits physical, 48 bits virtual  
Byte Order:            Little Endian  
CPU(s):                12  
On-line CPU(s) list:   0-11  
Vendor ID:             GenuineIntel  
Model name:            13th Gen Intel(R) Core(TM) i5-1335U  
CPU family:            6  
Model:                186  
Thread(s) per core:    2  
Core(s) per socket:    6  
Socket(s):             1  
Stepping:              3  
BogoMIPS:              4991.99  
Flags:                 fpu vme de pse tsc msr pae mce cx8  
                        apic sep mtrr pge mca cmov pat pse3  
                        6 clflush mmx fxsr sse sse2 ss ht s  
                        yscall nx pdpe1gb rdtscp lm constan  
                        t_tsc rep_good nopl xtopology tsc_r  
                        eliable nonstop_tsc cpuid tsc_known  
                        _freq pni pclmulqdq vmx ssse3 fma c  
                        x16 pcid sse4_1 sse4_2 x2apic movbe  
                        popcnt tsc_deadline_timer aes xsav  
                        e avx f16c rdrand hypervisor lahf_l  
                        m abm 3dnowprefetch ssbd ibrs ibpb  
                        stibp ibrs_enhanced tpr_shadow ept  
                        vpid ept_ad fsgsbase tsc_adjust bmi  
                        1 avx2 smep bmi2 erms invpcid rdsee  
                        d adx smap clflushopt clwb sha_ni x  
                        saveopt xsavec xgetbv1 xsaves avx_v  
                        nni vnmi umip waitpkg gfni vaes vpc  
                        lmulqdq rdpid movdiri movdir64b fsr  
                        m md_clear serialize ibt flush_lld  
                        arch_capabilities
```

```

Virtualization features:
  Virtualization:      VT-x
  Hypervisor vendor:   Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d:                 288 KiB (6 instances)
  L1i:                 192 KiB (6 instances)
  L2:                  7.5 MiB (6 instances)
  L3:                  12 MiB (1 instance)
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Not affected
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                 Not affected
  Meltdown:            Not affected
  Mmio stale data:      Not affected
  Old microcode:        Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:             Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:    Mitigation; Speculative Store Bypass
                        disabled via prctl
  Spectre v1:          Mitigation; usercopy/swapgs barriers
                        and __user pointer sanitization
  Spectre v2:          Mitigation; Enhanced / Automatic IBRS;
                        IBPB conditional; PBRSE-eIBRS
                        SW sequence; BHI BHI_DIS_S
  Srbds:               Not affected
  Tsa:                 Not affected
  Tsx async abort:     Not affected
  Vmscape:             Not affected

```

3.Ps

```

root@keerthana:~# ps
  PID TTY          TIME CMD
  545 pts/2    00:00:00 bash
  570 pts/2    00:00:00 ps
root@keerthana:~# ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1      0  0  04:59 ?        00:00:00 /sbin/init
root           2      1  0  04:59 hvcd    00:00:00 /init
root           6      2  0  04:59 hvcd    00:00:00 plan9 --control-socket 7 --log-level 4 --server-
root          52      1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd-journald
root         100      1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd-udevd
systemd+     111      1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     117      1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root         177      1  0  04:59 ?        00:00:00 /usr/sbin/cron -f -P
message+     180      1  0  04:59 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofo
root         188      1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         190      1  0  04:59 ?        00:00:00 /usr/libexec/ssl-pro-service -vv
root         202      1  0  04:59 tty1    00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog       209      1  0  04:59 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         219      1  0  04:59 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/
root         306      2  0  04:59 pts/1    00:00:00 /bin/login -f
root         349      1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd --user
root         350     349  0  04:59 ?        00:00:00 (sd-pam)
root         374     306  0  04:59 pts/1    00:00:00 -bash
root         501      2  0  05:12 ?        00:00:00 /init
root         503     501  0  05:12 ?        00:00:00 /init
root         509     503  0  05:12 pts/0    00:00:00 -bash
root         540      2  0  05:16 ?        00:00:00 /init
root         542     540  0  05:16 ?        00:00:00 /init
root         545     542  0  05:16 pts/2    00:00:00 -bash
root         571     545  0  05:18 pts/2    00:00:00 ps -ef
root@keerthana:~#

```

4.Top

```
root@keerthana:~# top
top - 05:19:49 up 20 min, 1 user, load average: 0.02, 0.02, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.1 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7796.6 total, 7207.1 free, 535.3 used, 203.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7261.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
190	root	20	0	1756104	13240	10884	S	0.3	0.2	0:00.16	wsl-pro-service
574	root	20	0	9316	5704	3504	R	0.3	0.1	0:00.01	top
1	root	20	0	21520	13092	9760	S	0.0	0.2	0:00.65	systemd
2	root	20	0	3180	2200	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	34056	12580	11504	S	0.0	0.2	0:00.21	systemd-journal
100	root	20	0	25524	6856	5168	S	0.0	0.1	0:00.30	systemd-udev
111	systemd+	20	0	21460	13192	10940	S	0.0	0.2	0:00.13	systemd-resolve
117	systemd+	20	0	91028	7992	7024	S	0.0	0.1	0:00.09	systemd-timesyn
177	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
180	message+	20	0	9596	5312	4628	S	0.0	0.1	0:00.08	dbus-daemon
188	root	20	0	17972	8832	7792	S	0.0	0.1	0:00.10	systemd-logind
202	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.01	agetty
209	syslog	20	0	222516	5936	4564	S	0.0	0.1	0:00.07	rsyslogd
219	root	20	0	107016	23240	13776	S	0.0	0.3	0:00.14	unattended-upgr
306	root	20	0	6676	4452	3680	S	0.0	0.1	0:00.01	login
349	root	20	0	20312	11636	9516	S	0.0	0.1	0:00.10	systemd
350	root	20	0	21164	3520	1804	S	0.0	0.0	0:00.00	(sd-pam)
374	root	20	0	6080	5284	3660	S	0.0	0.1	0:00.01	bash
501	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
503	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.03	Relay(509)
509	root	20	0	6076	5356	3672	S	0.0	0.1	0:00.05	bash
540	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
542	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.02	Relay(545)
545	root	20	0	6076	5336	3652	S	0.0	0.1	0:00.04	bash